

Blood Product Transfusion Reaction

Recognition, Immediate Management, Investigation & Reporting

Regulation: 42 CFR §482.27 | AABB Standards | TJC MM.05.01.09

PURPOSE & SCOPE

This policy establishes standardized procedures for the recognition, immediate management, laboratory investigation, documentation, and reporting of all suspected transfusion reactions. It applies to all clinical staff involved in blood product administration across all inpatient units, the emergency department, procedural areas, and the outpatient infusion center.

DEFINITIONS & REACTION TYPES

A transfusion reaction is any adverse event temporally associated with the transfusion of a blood product. Reactions may be acute (occurring during or within 24 hours of transfusion) or delayed (occurring more than 24 hours post-transfusion). All suspected reactions must be treated as potentially life-threatening until proven otherwise.

Reaction Type	Clinical Features & Immediate Action
Acute Hemolytic (AHTR)	ABO incompatibility; most fatal. Fever, chills, back/flank pain, hemoglobinuria, hypotension, DIC. Medical emergency — STOP transfusion immediately.
Febrile Non-Hemolytic (FNHTR)	Most common reaction. Temp rise $\geq 1^{\circ}\text{C}$ from baseline. Chills, rigors. No hemolysis. Stop transfusion; rule out AHTR before restarting.
Allergic / Urticarial	Hives, pruritus, flushing without systemic involvement. May slow or pause transfusion; administer antihistamine. Resume if symptoms resolve.
Anaphylaxis	Severe systemic reaction: bronchospasm, angioedema, hypotension, cardiovascular collapse. Stop transfusion immediately — treat as anaphylactic emergency.
Transfusion-Related Acute Lung Injury (TRALI)	New acute hypoxemia + bilateral pulmonary infiltrates within 6 hrs. Leading cause of transfusion-related mortality. Stop transfusion; supportive care; ICU.
Transfusion-Associated Circulatory Overload (TACO)	Acute pulmonary edema: hypertension, dyspnea, hypoxia, tachycardia. Risk: elderly, cardiac/renal disease. Stop/slow transfusion; diuretics; monitor.
Delayed Hemolytic (DHTR)	Occurs 24 hrs–30 days post-transfusion. Unexpected drop in Hgb, new antibody on repeat type & screen. Notify blood bank; monitor renal function.
Septic Transfusion Reaction	Bacterial contamination of product. High fever ($>2^{\circ}\text{C}$), rigors, shock. Stop immediately; blood cultures from patient and product; broad-spectrum antibiotics.

Transfusion-Associated Graft vs Host Disease (TA-GvHD)	Rare; donor T-cells attack host. Fever, rash, hepatitis, pancytopenia 1–6 weeks post. Prevention via irradiation of cellular products for at-risk patients.
Hypotensive Reaction	Isolated hypotension (SBP drop ≥ 30 mmHg) during transfusion without other signs. Often ACE inhibitor-related. Stop transfusion; evaluate for other causes.

■ CRITICAL SAFETY ALERT — STOP TRANSFUSION FOR ANY OF THE FOLLOWING

New fever ($\geq 1^\circ\text{C}$ rise from baseline) | Chills or rigors | Back, flank, or chest pain

Hypotension (SBP drop ≥ 30 mmHg or SBP < 90 mmHg) | Tachycardia (new or worsening)

Hemoglobinuria (red/brown urine) | Urticaria / angioedema | Respiratory distress / SpO₂ drop

Sense of impending doom reported by patient | Any other unexplained clinical change

IMMEDIATE MANAGEMENT — STEP-BY-STEP PROTOCOL

1 STOP the Transfusion Immediately

Clamp the blood tubing at the IV site. Do NOT remove the IV access. Do NOT discard the blood product or tubing — both are required for the reaction workup.

2 Maintain IV Access with Normal Saline

Keep the IV patent by infusing 0.9% Normal Saline (NS) through NEW IV tubing connected directly to the IV catheter. Do NOT flush blood product into the patient using existing tubing.

3 Perform Rapid Patient Assessment

Obtain vital signs (temperature, BP, HR, RR, SpO₂). Perform focused assessment: skin (rash, flushing, hives), respiratory (wheezing, distress), renal (urine color), neurological (LOC, complaint of pain). Assign severity: mild / moderate / severe.

4 Notify Physician and Blood Bank Immediately

Contact the attending physician or rapid response team (for moderate/severe reactions). Simultaneously notify the Blood Bank of the suspected reaction. Provide: unit number, patient name/MRN, time transfusion started, time reaction noted, current vital signs, and symptoms.

5 Collect Post-Reaction Specimens

Draw the following BEFORE any additional IV fluids are given when possible: • 1 pink-top (EDTA) tube — repeat type & crossmatch, DAT • 1 red-top (SST) tube — plasma for hemolysis evaluation (BMP, LDH, haptoglobin, bilirubin) • Urine specimen — free hemoglobin (hemoglobinuria) • Blood cultures $\times 2$ sets if septic reaction is suspected Label specimens with TWO patient identifiers and mark clearly as 'TRANSFUSION REACTION WORKUP.'

6	Return Blood Product and Tubing to Blood Bank Return the implicated blood bag (with all attached tubing, intact and clamped), the transfusion record, and any IV solutions co-infused. The blood bank will inspect the unit and perform a clerical check.
7	Administer Treatment as Ordered Treatment is reaction-specific and physician-directed: • Allergic/urticarial: Diphenhydramine 25–50 mg IV/IM per order • Anaphylaxis: Epinephrine 0.3–0.5 mg IM (1:1,000) immediately; call Code/RRT • AHTR/Hemolytic: IV fluids to maintain UO ≥1 mL/kg/hr; furosemide per order; monitor renal function • TACO: Furosemide 20–40 mg IV; upright positioning; supplemental O ₂ • Febrile (FNHTR confirmed): Acetaminophen 650 mg PO/PR per order • Septic: Blood cultures × 2; broad-spectrum antibiotics per order; notify pharmacy and ID
8	Document in the Medical Record Complete nursing documentation must include: time transfusion began, time reaction noted, vital signs trend (pre/during/post), symptoms described by the patient and observed, all interventions performed with times, name and time of physician notification, patient response to treatment, and disposition of blood product.
9	Complete Transfusion Reaction Report Complete the facility transfusion reaction report form within 1 hour of reaction identification. Submit to the Blood Bank AND enter a Patient Safety Event (incident) report in the occurrence reporting system. Do not delay reporting pending lab results.
10	Monitor Patient Post-Reaction Vital signs and assessment every 15 minutes × 4, then every 30 minutes × 2, then per unit protocol. Monitor urine output. If transfusion is to be resumed (mild allergic only, after physician order and symptom resolution), restart at ≤1 mL/kg/hr with bedside nurse present for first 15 minutes.

CRITERIA FOR RESUMING TRANSFUSION

Transfusion may ONLY be resumed under the following conditions and requires an explicit physician order:

Reaction Type	Resume Criteria
Mild urticarial (hives only)	May resume after antihistamine administered and symptoms fully resolved. Physician order required.
Febrile Non-Hemolytic (FNHTR confirmed)	May resume same unit after acetaminophen and symptom resolution IF hemolytic reaction has been ruled out by blood bank clerical check. Physician order required.
AHTR / Hemolytic	NEVER resume the implicated unit. Transfuse new, fully crossmatched unit only after investigation complete and physician/blood bank authorization.
Anaphylaxis / TRALI / Septic	NEVER resume the implicated unit. Discuss with blood bank medical director before any further transfusion.
TACO	May consider resuming at a slower rate after diuresis and symptom resolution at physician discretion with close monitoring.

SPECIAL POPULATIONS & CONSIDERATIONS

Patients Who Decline Blood Products (Religious/Personal Objection)

Patients who have declined blood products in advance (e.g., Jehovah's Witnesses with signed refusal) must have blood product refusal clearly flagged in the medical record and on the armband (per facility policy). In the event of a transfusion reaction in a patient who received blood despite refusal, all standard reaction management applies. Ethics and legal counsel must be notified immediately. Document all events meticulously.

Pediatric Patients

Reaction thresholds in pediatric patients: fever is defined as temperature $\geq 38.5^{\circ}\text{C}$ or any rise $\geq 1^{\circ}\text{C}$ above pre-transfusion baseline. Volume thresholds for TACO are weight-based. Notify pediatric attending and/or PICU for any moderate/severe reaction. Pediatric dosing for treatment medications must be verified with pharmacy before administration.

Massive Transfusion Protocol (MTP) Patients

Patients receiving uncrossmatched emergency-release blood are at increased risk of alloimmune reactions when subsequently crossmatched products become available. All MTP patients must have a post-transfusion type and screen performed after stabilization. Reaction thresholds may be masked by underlying hemodynamic instability — any unexplained clinical deterioration should prompt consideration of a transfusion reaction.

REPORTING, INVESTIGATION & QUALITY REVIEW

Mandatory Reporting Requirements

- All transfusion reactions must be reported to the Blood Bank within 1 hour of recognition.
- Serious adverse events (hemolytic, TRALI, septic, fatal reactions) must be reported to the FDA on Form 3500A (MedWatch) within 7 days for serious reactions, or 15 days for fatal reactions.
- Facilities accredited by AABB must report transfusion fatalities to AABB within 24 hours.
- State reporting requirements vary — refer to the current State Health Department reporting matrix.
- All reaction events must be entered in the facility Patient Safety Event Reporting System.

The Transfusion Committee (or Blood Utilization Review Committee) must review all transfusion reactions at least quarterly. The blood bank medical director must review all serious reactions within 24 hours. Root cause analysis is required for all TRALI events, fatal reactions, and hemolytic reactions due to clerical error. Findings and action plans must be reported to the QAPI program.

DOCUMENTATION CHECKLIST

Documentation Element	Status	Notes
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Pre-transfusion vitals	Required	Baseline within 30 min before hang
Two-person ID verification	Required	Both names documented on transfusion record
15-minute assessment & vitals	Required	Nurse present at bedside
Reaction onset time	Required	Exact time, not estimated
Symptoms reported/observed	Required	Verbatim patient statements + objective findings
Transfusion stop time	Required	Document immediately upon stopping
Physician notification time	Required	Name of physician notified
Blood bank notification time	Required	Name of blood bank staff notified
Specimens collected	Required	Type, time, labels verified
Treatment administered	Required	Drug, dose, route, time, response
Post-reaction vitals	Required	Q15 min × 4, then Q30 min × 2
Incident report filed	Required	Within 1 hour of reaction identification
Patient/family education	Required	Document teaching provided and understanding

For internal clinical use only. Review annually and upon regulatory updates. This policy does not replace clinical judgment.

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